WNetTech Interface Development & Evaluation Kit - Quick Start Manual
(For WNetTool & Initial Hardware Evaluation)

IMPORTANT NOTICE: This manual and all hardware models covered herein are intended solely for **Engineering Evaluation and Secondary Development**. This module is a **sub-assembly/component** and is not intended for direct sale to end consumers. Developers are responsible for obtaining the necessary compliance certifications (e.g., FCC/ISED) for their final integrated products.

---

1. Development Kit Evaluation Guide

This guide is designed to help engineers quickly verify the transparent transmission functionality of the ESP32-C3-MINI-1-N4 core evaluation board, facilitating data bridging demonstrations between industrial fieldbuses (CAN, RS485, SPI, UART) and wireless networks.

Computing Core: ESP32-C3-MINI-1-N4 (RISC-V @160MHz), integrated 4MB Flash and 2.4GHz WiFi.

Compliance Endorsement: This kit integrates a certified wireless module (FCC ID: 2AC7Z-ESP32C3MINI1).

Platform Desig: Unified wireless engine with dedicated signal processing circuits matched for different buses.

Physical Layer Adaptation:

- CAN Module: Integrates TCAN3414DR, supporting ISO 11898-2 standards.
- RS485 Module: Integrates 3.3V differential transceiver, supports hardware-level automatic direction switching.
- SPI/UART Module: Compatible with 3.3V and 5V data levels, no level conversion required.

Communication Modes: Supports AP mode (device configuration) and STA mode (network transmission).

2. Module Specifications & Interface Definitions

Module Model	Physical Layer Chip	Key Pin Assignment (ESP32-C3)	Interface Features
C3-CAN	TCAN3414DR	TX: GPIO2 / RX: GPIO3	Supports CAN 2.0B, max 1Mbps, strong ESD protection
C3-485	SP3485EN-L_TR	TX: GPIO4 / RX: GPIO5 / DE: GPIO6	Supports Modbus RTU transparent transmission
C3-UART	TTL Header	TX: GPIO4 / RX: GPIO5	Default: 115200, 8, N, 1
C3-SPI	TTL Header	SCK/MOSI/MISO/CS/REQ: GPIO 4,5,6,7,10	Module operates in **Slave Mode**

Note: Please confirm the signal level of the target system before wiring. For the C3-CAN module, please ensure the external power supply is a stable 5V. Improper connection may lead to damage to the development board or your host equipment.

3. LED Indicator Status Description

Monitor the board's LED to determine its current operational status:

- **1s Slow Flash (1s)**: The device is in **AP Configuration Mode**.
- **200ms Fast Flash**: **WiFi connected** successfully.
- **500ms Medium Flash**: Server connection successful, entered data transparent transmission ready state.

4. Parameter Configuration Tool Usage (WNetTool)

Two configuration schemes are provided for rapid evaluation:

Scheme A: Web Portal Configuration (AP Mode)

1. **Enter Configuration Mode**:
   - **Automatic Entry**: After power-on, if no WiFi is connected, the LED flashes slowly every 1 second, entering AP mode.

- **Forced Entry**: Press and hold the **Config** button for more than 3 seconds until the LED flashes every 1 second.

2. **Connect to Hotspot**: Use a phone or computer to connect to the hotspot named SmartUART_Config (or corresponding protocol name), Password: 12345678.
3. **Access Backend**: Open a browser and navigate to **192.168.8.1** for SSID, Server IP, and Baudrate settings.
4. **Set Protocol Parameters**:
   - **SSID**: WiFi name.
   - **Password**: WiFi password.
   - **Server IP**: Server IP address (e.g., 192.168.2.18).
   - **Port**: Server target listening port (e.g., 8888).
   - **Baudrate**: Baud rate (e.g., 115200), Data bits (appears only for UART/RS485 settings).
5. **Save & Apply**: Click **Save & Connect** to send the configuration.

Scheme B: USB Setting Tool (For Batch Evaluation)

1. Download Tool: https://github.com/wnettech/WNetTech-Tool

Download Parameter Setting Software:

Chinese Version: WNetTool_Config_v1.0.0_cn.exe

English Version: WNetTool_Config_v1.0.0_en.exe

2. Environment Preparation: It is recommended to run the tool in an English path such as C:\My_Project_Tools.

3. Functionality: Supports quick distribution of network configuration and bus parameters via USB serial port.

4. Protocol Parameter Settings:

o SSID: WiFi name.

o Password: WiFi password.

- Server IP: Server IP address (e.g., 192.168.2.18).
- Port: Server target listening port (e.g., 8888).
- Baudrate: Baud rate (e.g., 115200), data bits (only appears when setting UART/RS485).

5. Transparent Data Transmission Test

- **Prepare the server:** Download the server testing software from https://github.com/wnettech/WNetTech-Tool:
  Chinese version: WNetTool_Server_Port55001_v1.0.0_cn.exe
  English version: WNetTool_Server_Port55001_v1.0.0_en.exe
- **Obtain your local IP address:**
  Type `ipconfig` in the command line to obtain your IPv4 address (e.g., 192.168.18.4).
- **Configuration Setup:**
  Settings: WiFi Username and Password
  Set the server IP to your local machine's IP address, and the port to 55001 (as specified by the testing software).
  Baud rate (if it's a UART or RS485 module).

  After successful setup, the device will enter a 200ms fast flash (if the WiFi settings are correct).

- **Waiting to Connect to the Server**

  Open the testing transparency software:
  WNetTool_Server_Port55001_v1.0.0_cn.exe or
  WNetTool_Server_Port55001_v1.0.0_en.exe
  The device will change from a 200ms fast flash to a medium flash of 500ms, indicating that the device has successfully connected to the server.

- Data Testing

  Taking a UART module as an example: Open a serial port assistant (baud rate must be consistent), send the string "Hello WiFi". The PC testing software should be able to receive the raw data instantly in the test software's receive message.

Connect to the server.

- **Wiring Notes:** During the evaluation process, ensure correct signal cable wiring, especially the polarity of CAN_H/L and RS485_A/B.

- **Compliance Maintenance**: The WNetTool provided in this manual is for parameter configuration and functional verification only. For complete compliance documentation (SDoC, series declarations, etc.), please visit: https://github.com/wnettech/compliance

6. Technical Support and Documentation

- Manufacturer: WNetTech

- Compliance Documentation (SDoC/Declaration): https://github.com/wnettech/compliance

- Tool Download: https://github.com/wnettech/WNetTech-Tool This development kit is designed to comply with the technical requirements of FCC Part 15B and ICES-003 Class B. All models use a certified wireless module (FCC ID: 2AC7Z-ESP32C3MINI1) and no changes have been made to the RF circuitry.